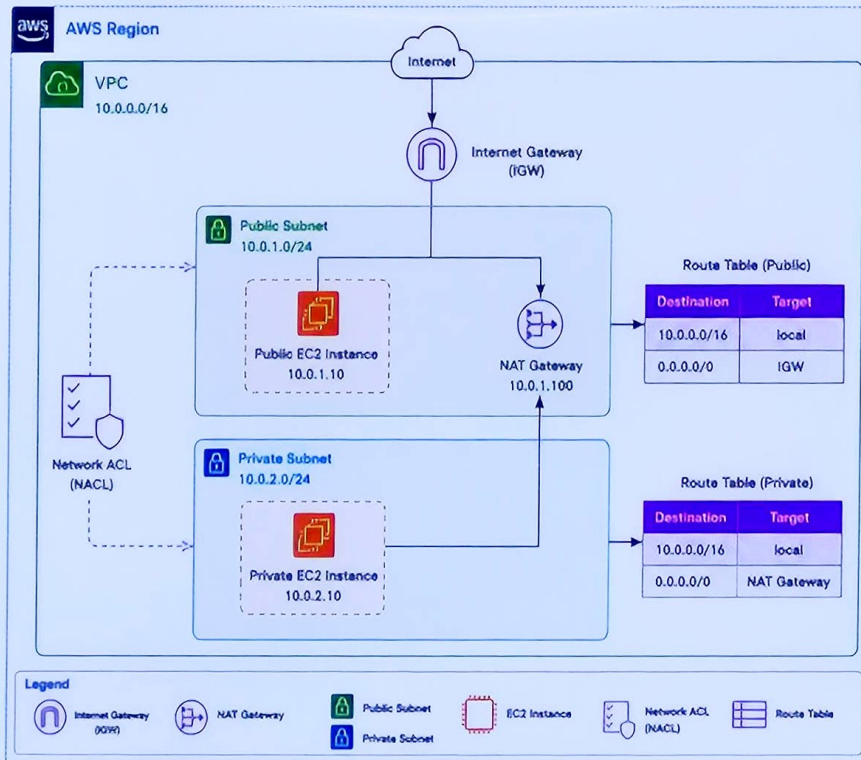


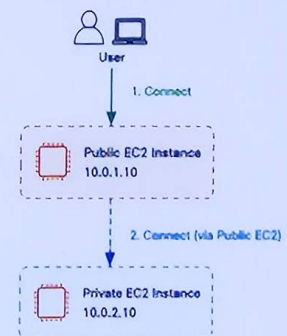


Key Points:

- Public Subnet has route to IGW for direct Internet access
- Private Subnet uses NAT Gateway for outbound Internet access
- NACL is associated with the subnet and controls inbound/outbound traffic
- Instances in private subnet are not directly accessible from Internet

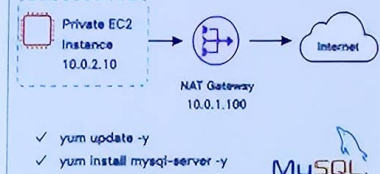
SCENARIO 1

1. Connect to Public EC2 Instance from Internet
2. Connect to Private EC2 Instance via Public EC2



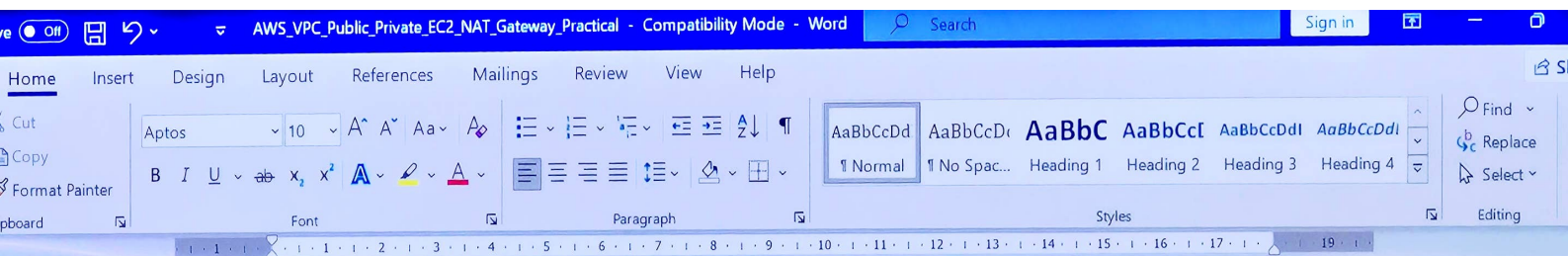
SCENARIO 2

1. NAT Gateway allows Private EC2 to access Internet
2. Update OS and Install MySQL in Private EC2



Private instance can access Internet via NAT Gateway but is NOT directly accessible from Internet

Reference architecture for the practical



5. Scenario 1 — Build VPC, Public/Private Subnets and EC2 Connectivity

Step 1 — Create the VPC

1. Open AWS Management Console → VPC.
2. Select Your VPCs → Create VPC.
3. Choose VPC only.
4. Name tag: Training-VPC.
5. IPv4 CIDR: 10.0.0.0/16.
6. IPv6 CIDR: No IPv6 CIDR block.
7. Tenancy: Default.
8. Click Create VPC.

Expected result: VPC Training-VPC with CIDR 10.0.0.0/16.

Step 2 — Create the Public Subnet

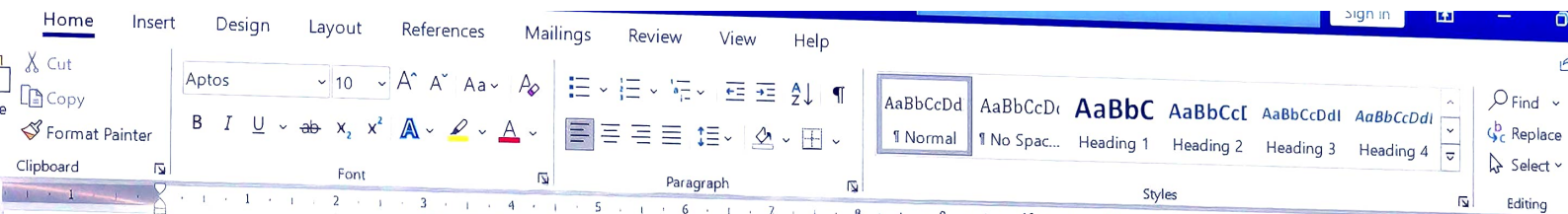
9. VPC Console → Subnets → Create subnet.
10. VPC: Training-VPC.
11. Subnet name: Public-Subnet.
12. Availability Zone: choose one AZ, for example ap-south-1a.



Step 2 — Create the Public Subnet

9. VPC Console → Subnets → Create subnet.
10. VPC: Training-VPC.
11. Subnet name: Public-Subnet.
12. Availability Zone: choose one AZ, for example ap-south-1a.
13. IPv4 subnet CIDR: 10.0.1.0/24.
14. Create subnet.

15. Select Public-Subnet → Actions → Edit subnet settings.
16. Enable 'Auto-assign public IPv4 address'. Save.



15. Select Public-Subnet → Actions → Edit subnet settings.

16. Enable 'Auto-assign public IPv4 address'. Save.

Step 3 — Create the Private Subnet

17. Create another subnet in the same VPC.

18. Subnet name: Private-Subnet.

19. Use the same AZ for a simple classroom demonstration, or another AZ if you want to demonstrate multi-AZ.

20. IPv4 subnet CIDR: 10.0.2.0/24.

21. Create subnet.

22. Do not enable auto-assign public IPv4 address.

Step 4 — Create and Attach the Internet Gateway

23. VPC Console → Internet Gateways → Create internet gateway.

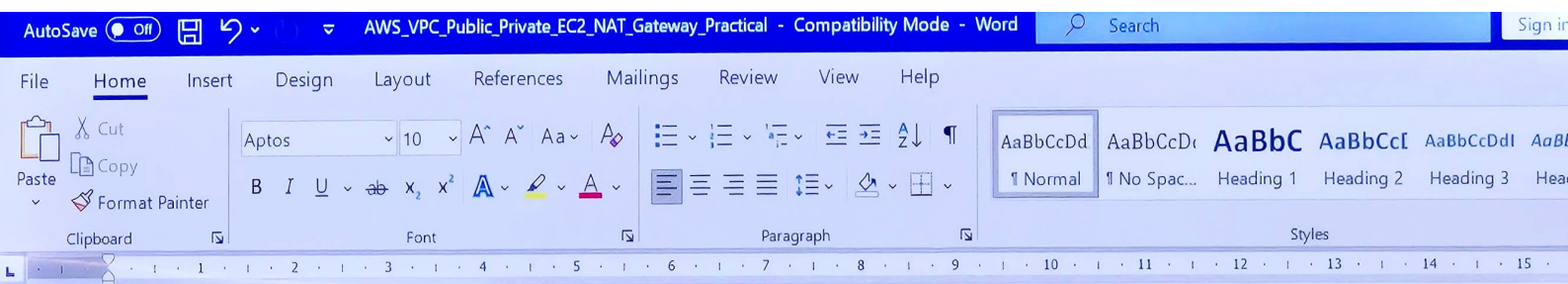
24. Name: Training-IGW.

25. Create internet gateway.

26. Select Training-IGW → Actions → Attach to VPC.

27. Select Training-VPC → Attach.





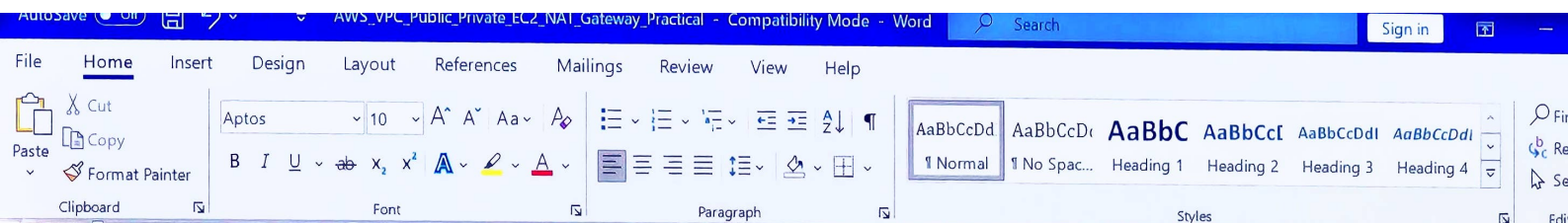
Step 4 — Create and Attach the Internet Gateway

23. VPC Console → Internet Gateways → Create internet gateway.
24. Name: Training-IGW.
25. Create internet gateway.
26. Select Training-IGW → Actions → Attach to VPC.
27. Select Training-VPC → Attach.

The Internet Gateway is required for direct Internet connectivity from resources in the public subnet.

Step 5 — Create the Public Route Table

28. VPC Console → Route Tables → Create route table.
29. Name: Public-RT.
30. VPC: Training-VPC.
31. Create.
32. Open Public-RT → Routes → Edit routes → Add route.
33. Destination: 0.0.0.0/0.



The internet Gateway is required for direct internet connectivity from resources in the public subnet.

Step 5 — Create the Public Route Table

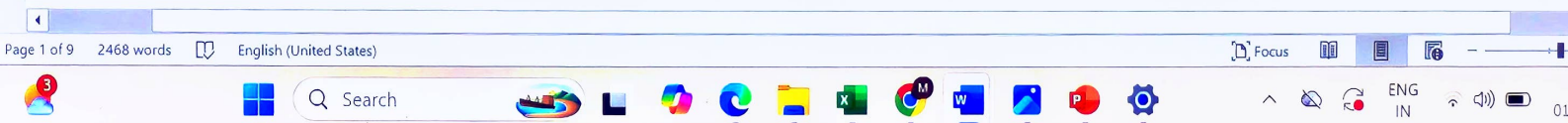
28. VPC Console → Route Tables → Create route table.
29. Name: Public-RT.
30. VPC: Training-VPC.
31. Create.
32. Open Public-RT → Routes → Edit routes → Add route.
33. Destination: 0.0.0.0/0.
34. Target: Internet Gateway → Training-IGW.
35. Save changes.
36. Open Subnet associations → Edit subnet associations → select Public-Subnet → Save.

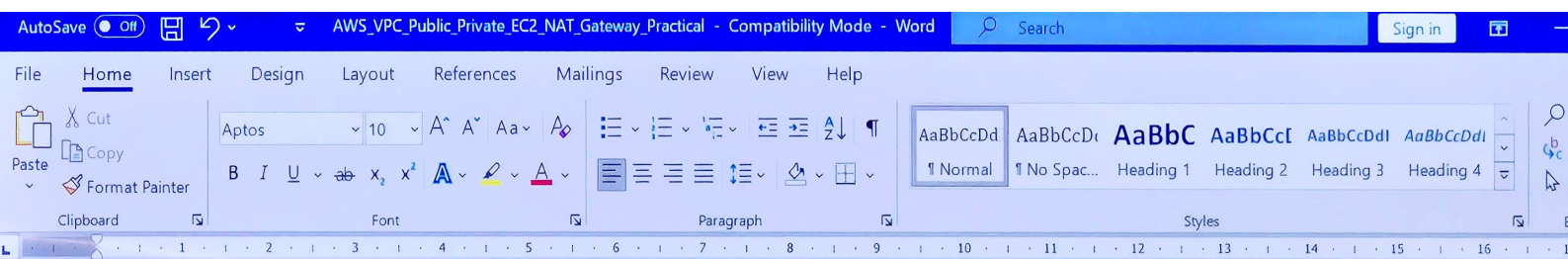
Public-RT:

10.0.0.0/16 → local

0.0.0.0/0 → Internet Gateway

Step 6 — Create the Private Route Table





Step 6 — Create the Private Route Table

37. VPC Console → Route Tables → Create route table.
38. Name: Private-RT.
39. VPC: Training-VPC.
40. Create.
41. Open Private-RT → Subnet associations → Edit subnet associations → select Private-Subnet → Save.
42. Do not add an Internet Gateway route yet. This is intentional for Scenario 1.

Private-RT before NAT:

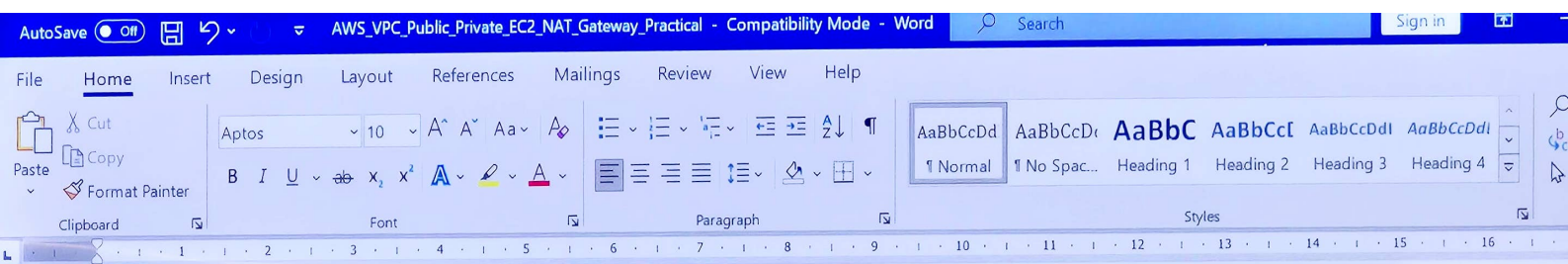
10.0.0.0/16 → local

(no 0.0.0.0/0 Internet route)

Step 7 — Create the Custom Network ACL

A Network ACL is associated with a subnet and is stateless. use a simple custom NACL that allows the traffic required for the demonstration. Remember that security groups are stateful, while NACLs are stateless.

43. VPC Console → Network ACLs → Create network ACL.



Step 7 — Create the Custom Network ACL

A Network ACL is associated with a subnet and is stateless. use a simple custom NACL that allows the traffic required for the demonstration. Remember that security groups are stateful, while NACLs are stateless.

43. VPC Console → Network ACLs → Create network ACL.

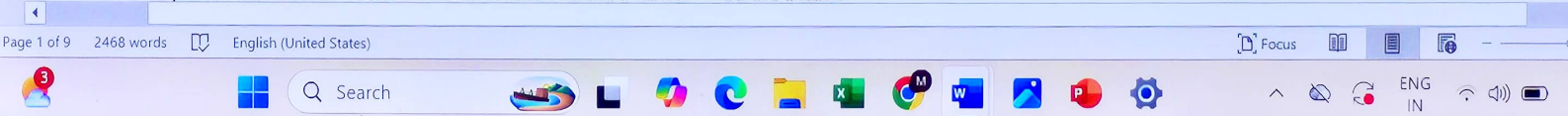
44. Name: Training-NACL.

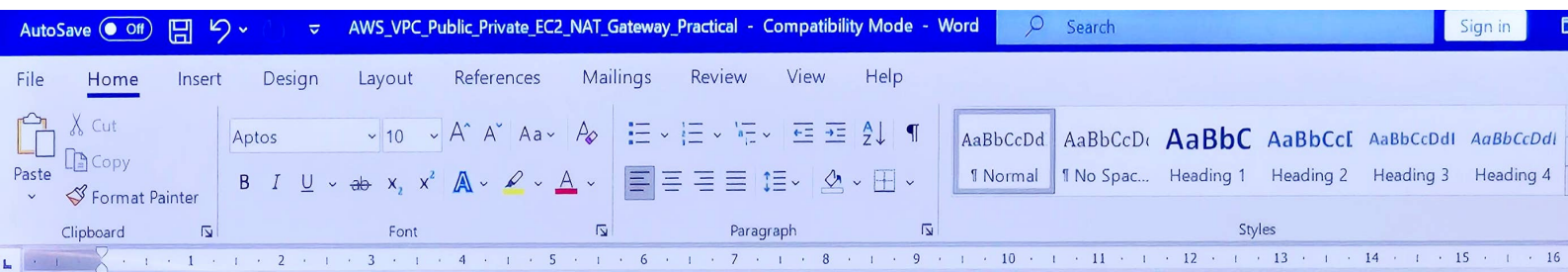
I

45. VPC: Training-VPC.

46. Create.

47. Open Inbound rules → Edit inbound rules → Add rule.





45. VPC: Training-VPC.

46. Create.

47. Open Inbound rules → Edit inbound rules → Add rule.

48. Rule 100: Type ALL traffic, Source 0.0.0.0/0, Allow.

49. Open Outbound rules → Edit outbound rules → Add rule.

50. Rule 100: Type ALL traffic, Destination 0.0.0.0/0, Allow.

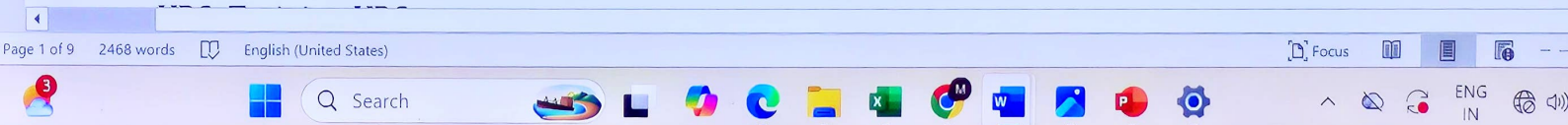
51. Open Subnet associations → Edit → associate Public-Subnet and Private-Subnet.

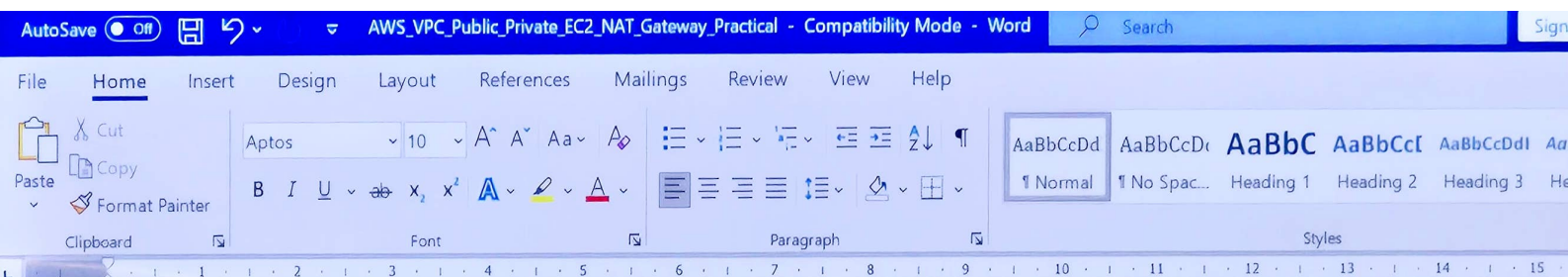
For production, do not blindly allow all traffic. Tighten NACL rules according to the application's ports and source/destination requirements.

Step 8 — Create Security Groups

Create two security groups so that the demonstration clearly shows the difference between public and private

8A. Public-EC2-SG





source/destination requirements.

Step 8 — Create Security Groups

Create two security groups so that the demonstration clearly shows the difference between public and

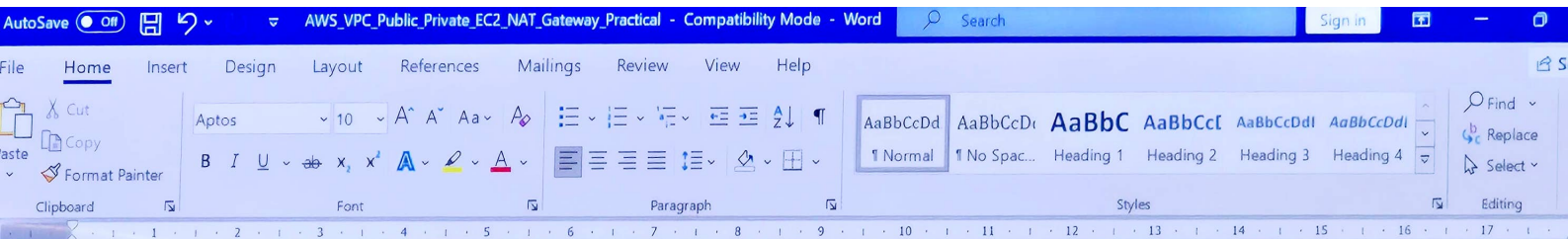
8A. Public-EC2-SG

- VPC: Training-VPC.
- Inbound SSH TCP 22: Source = My IP (recommended for the laptop used for SSH).
- Optional ICMP rule may be added for ping testing, but it is not required.
- Outbound: allow required outbound traffic. The default outbound allow rule is acceptable for the lab.

8B. Private-EC2-SG

- VPC: Training-VPC.
- Inbound SSH TCP 22: Source = Public-EC2-SG (select the security group as the source, not 0.0.0.0/0).
- Outbound: allow required outbound traffic. Default outbound allow is acceptable for the lab.

This is the preferred design: the private instance accepts SSH only from the public EC2 security group.



source/destination requirements.

Step 8 — Create Security Groups

Create two security groups so that the demonstration clearly shows the difference between public and private access.

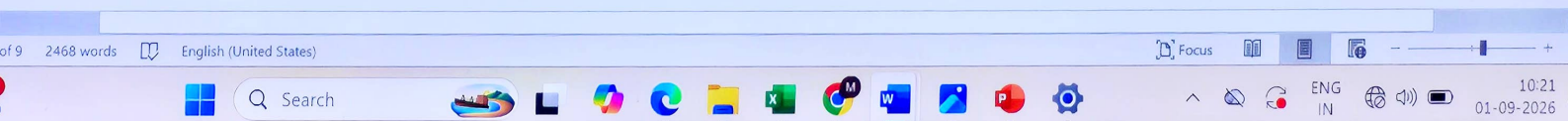
8A. Public-EC2-SG

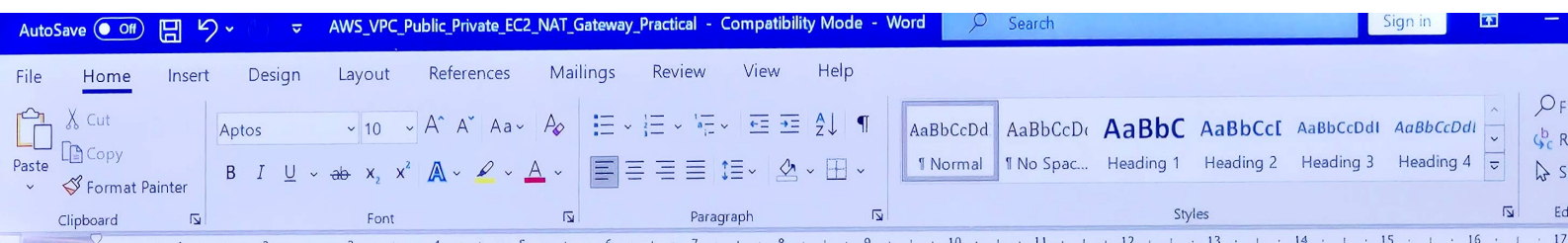
- VPC: Training-VPC.
- Inbound SSH TCP 22: Source = My IP (recommended for the laptop used for SSH).
- Optional ICMP rule may be added for ping testing, but it is not required.
- Outbound: allow required outbound traffic. The default outbound allow rule is acceptable for the lab.

8B. Private-EC2-SG

- VPC: Training-VPC.
- Inbound SSH TCP 22: Source = Public-EC2-SG (select the security group as the source, not 0.0.0.0/0).
- Outbound: allow required outbound traffic. Default outbound allow is acceptable for the lab.

This is the preferred design: the private instance accepts SSH only from the public EC2 security group.





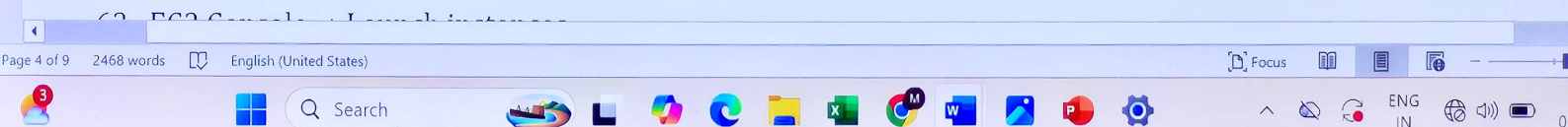
This is the preferred design: the private instance accepts SSH only from the public EC2 security group.

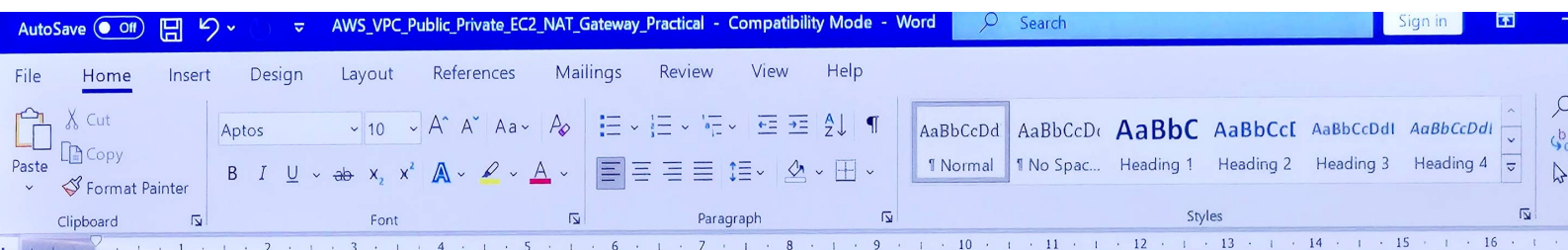
Step 9 — Create the Public EC2 Instance

52. Open EC2 Console → Instances → Launch instances.
53. Name: Public-EC2.
54. Choose an Amazon Linux AMI.
55. Instance type: choose a small/free-tier-eligible type available to your account.
56. Select or create an SSH key pair, for example Training-Key.
57. Network settings → VPC: Training-VPC.
58. Subnet: Public-Subnet.
59. Auto-assign Public IP: Enable.
60. Security group: Public-EC2-SG.
61. Launch instance.

After the instance becomes Running, record its Public IPv4 address and Private IPv4 address.

Step 10 — Create the Private EC2 Instance

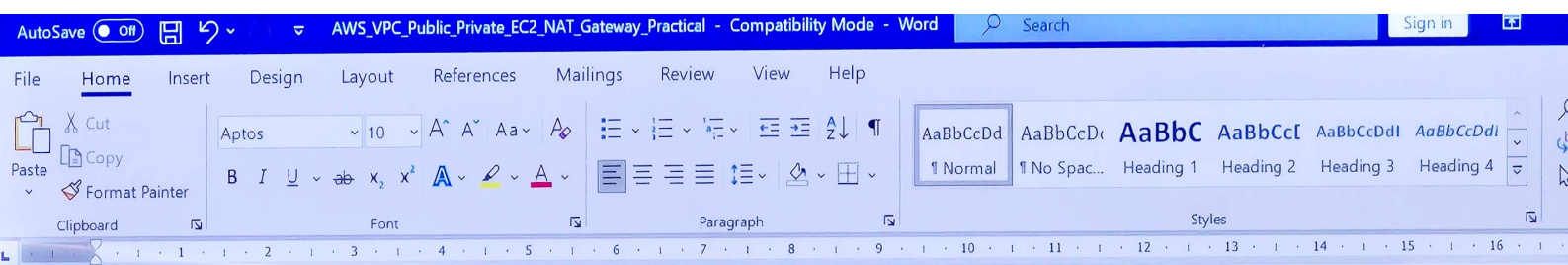




After the instance becomes Running, record its Public IPv4 address and Private IPv4 address.

Step 10 — Create the Private EC2 Instance

62. EC2 Console → Launch instances.
63. Name: Private-EC2.
64. Use the same AMI family as the public instance.
65. Use the same key pair, Training-Key, for easier SSH demonstration.
66. VPC: Training-VPC.
67. Subnet: Private-Subnet.
68. Auto-assign Public IP: Disable.
69. Security group: Private-EC2-SG.
70. Launch instance.



Record the private IPv4 address. There should be no public IPv4 address.

Step 11 — Test Public EC2 Connectivity from Your Laptop

On your local machine, connect to the public instance using its public IP.

```
chmod 400 Training-Key.pem
ssh -i Training-Key.pem ec2-user@<PUBLIC_EC2_PUBLIC_IP>
```

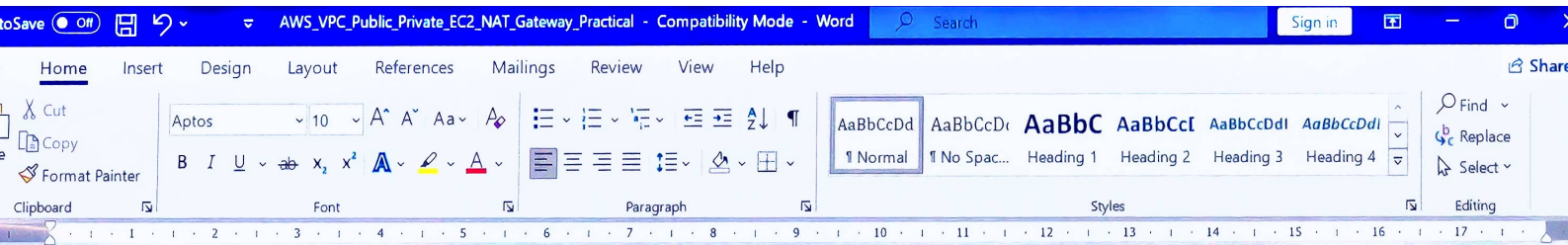
If using Windows PowerShell with OpenSSH, the `chmod` command may not be required. Use the SSH command directly after securing the key file permissions.

Step 12 — From Public EC2, Connect to Private EC2

Copy the private EC2 private IP and key file to the public instance only for this temporary classroom demonstration. A better production approach is SSH agent forwarding or AWS Systems Manager Session Manager.

```
scp -i Training-Key.pem Training-Key.pem ec2-user@<PUBLIC_EC2_PUBLIC_IP>:/home/ec2-user/
ssh -i Training-Key.pem ec2-user@<PRIVATE_EC2_PRIVATE_IP>
```

If the SCP command is run from your laptop, the first command uploads the key. Then, after SSHing to Public-EC2,



after securing the key file permissions.

Step 12 — From Public EC2, Connect to Private EC2

Copy the private EC2 private IP and key file to the public instance only for this temporary classroom demonstration. A better production approach is SSH agent forwarding or AWS Systems Manager Session Manager.

```
scp -i Training-Key.pem Training-Key.pem ec2-user@<PUBLIC_EC2_PUBLIC_IP>:/home/ec2-user/  
ssh -i Training-Key.pem ec2-user@<PRIVATE_EC2_PRIVATE_IP>
```

If the SCP command is run from your laptop, the first command uploads the key. Then, after SSHing to Public-EC2, run:

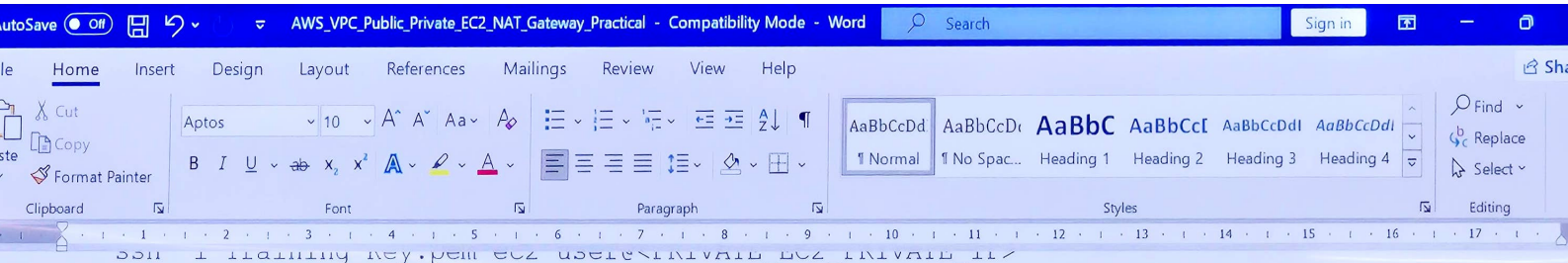
```
chmod 400 Training-Key.pem  
ssh -i Training-Key.pem ec2-user@<PRIVATE_EC2_PRIVATE_IP>
```

Expected result: you should reach the shell of Private-EC2 even though Private-EC2 has no public IP. The connection path is Laptop → Public EC2 → Private EC2.

Step 13 — Verify the Private Instance Has No Direct Internet Access

On Private-EC2, try:





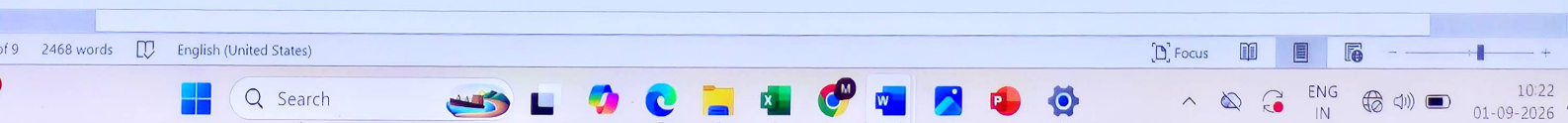
Expected result: you should reach the shell of Private-EC2 even though Private-EC2 has no public IP. The connection path is Laptop → Public EC2 → Private EC2.

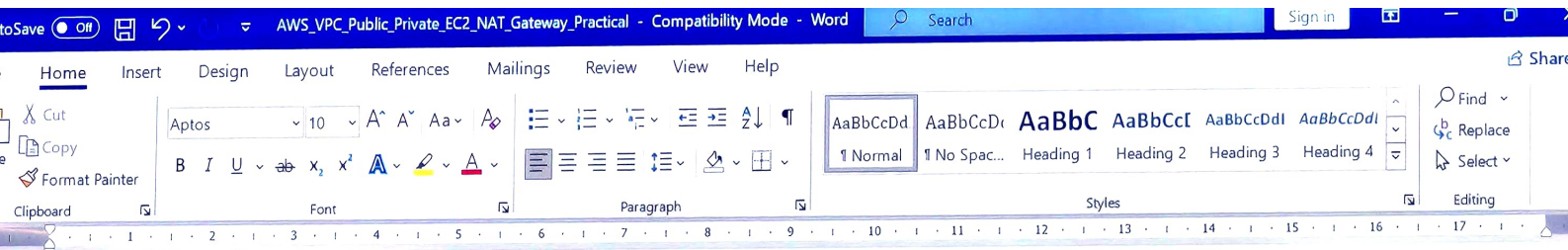
Step 13 — Verify the Private Instance Has No Direct Internet Access

On Private-EC2, try:

```
curl -I https://www.google.com
```

Before NAT Gateway is configured, Internet access should fail or time out because Private-RT has no default route to an Internet Gateway.





6. Scenario 2 — Add NAT Gateway and Enable Private EC2 Outbound Internet

A NAT Gateway allows instances in a private subnet to initiate outbound connections to the Internet. The private instance does not receive a public IPv4 address, and Internet hosts cannot initiate a new inbound connection to the private instance through the NAT Gateway.

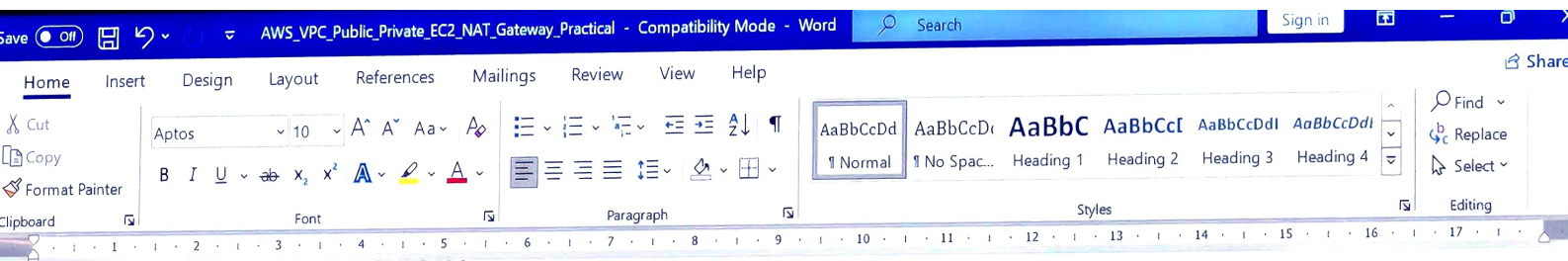
I

Step 14 — Allocate an Elastic IP for the NAT Gateway

71. VPC Console → Elastic IPs.
72. Click Allocate Elastic IP address.
73. Network border group: keep the Region default.
74. Click Allocate.
75. Record the allocated Elastic IP.

Step 15 — Create the NAT Gateway in the Public Subnet





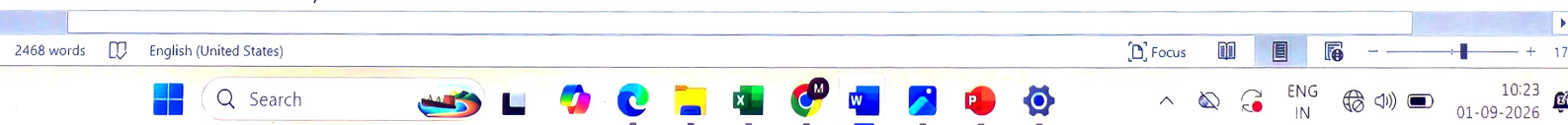
Step 15 — Create the NAT Gateway in the Public Subnet

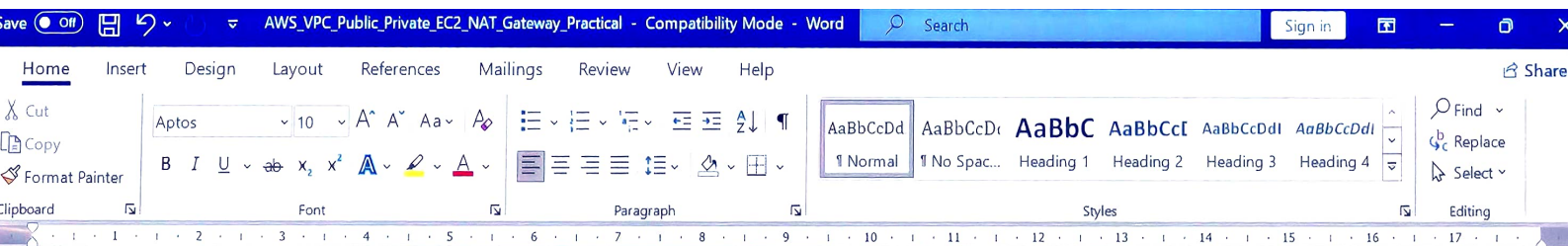
76. VPC Console → NAT Gateways → Create NAT gateway.
77. Subnet: Public-Subnet.
78. Connectivity type: Public.
79. Elastic IP allocation ID: select the Elastic IP created in the previous step.
80. Name: Training-NAT-GW.
81. Create NAT gateway.
82. Wait until the NAT Gateway state becomes Available.

Important: A public NAT Gateway must be placed in a subnet whose route table sends Internet-bound traffic to an Internet Gateway.

Step 16 — Update the Private Route Table

83. VPC Console → Route Tables → select Private-RT.
84. Routes → Edit routes → Add route.
85. Destination: 0.0.0.0/0.





Important: A public NAT Gateway must be placed in a subnet whose route table sends internet-bound traffic to an Internet Gateway.

Step 16 — Update the Private Route Table

83. VPC Console → Route Tables → select Private-RT.

84. Routes → Edit routes → Add route.

85. Destination: 0.0.0.0/0.

86. Target: NAT Gateway → Training-NAT-GW.

87. Save changes.

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Private-RT after NAT:

10.0.0.0/16 → local

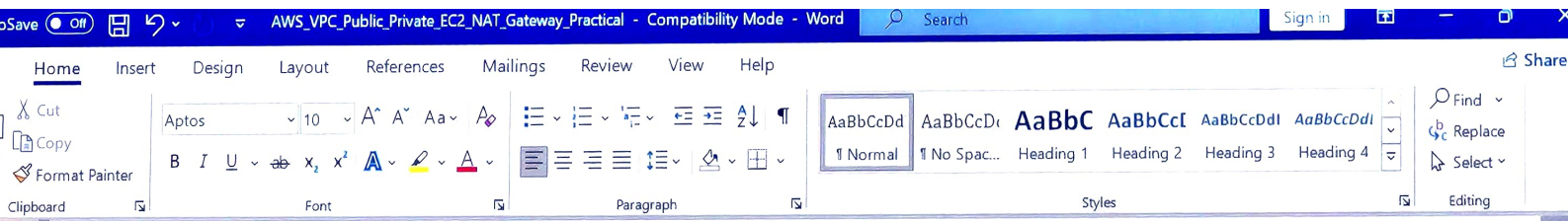
0.0.0.0/0 → NAT Gateway

The complete outbound path is: Private EC2 → Private Route Table → NAT Gateway → Public Route Table → Internet Gateway → Internet.

Step 17 — Test Internet Access from Private EC2

```
curl -I https://www.google.com
```





The complete outbound path is: Private EC2 → Private Route Table → NAT Gateway → Public Route Table → Internet Gateway → Internet.

Step 17 — Test Internet Access from Private EC2

```
curl -I https://www.google.com
```

You can also test DNS:

```
nslookup google.com
```

If the tests work, the private instance now has outbound Internet access through the NAT Gateway.

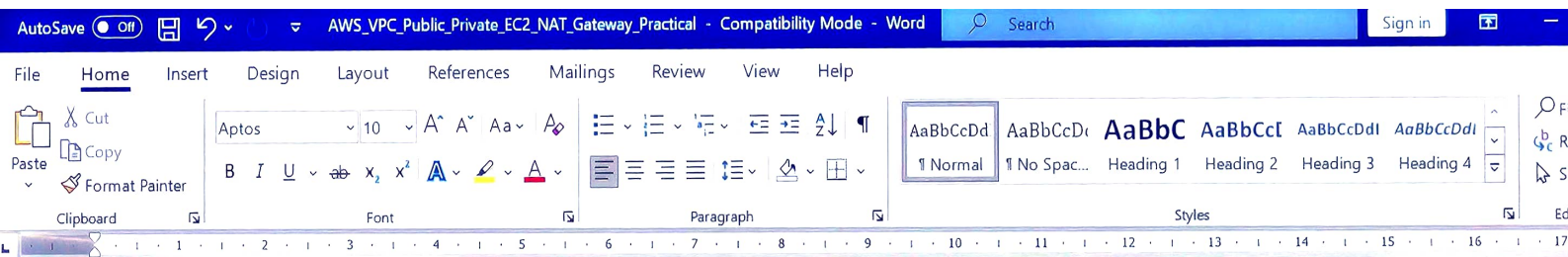
Step 18 — Update the Operating System

For Amazon Linux 2023, run:

```
sudo dnf update -y
```

For Amazon Linux 2, run:





```
sudo dnf update -y
```

For Amazon Linux 2, run:

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```
sudo yum update -y
```

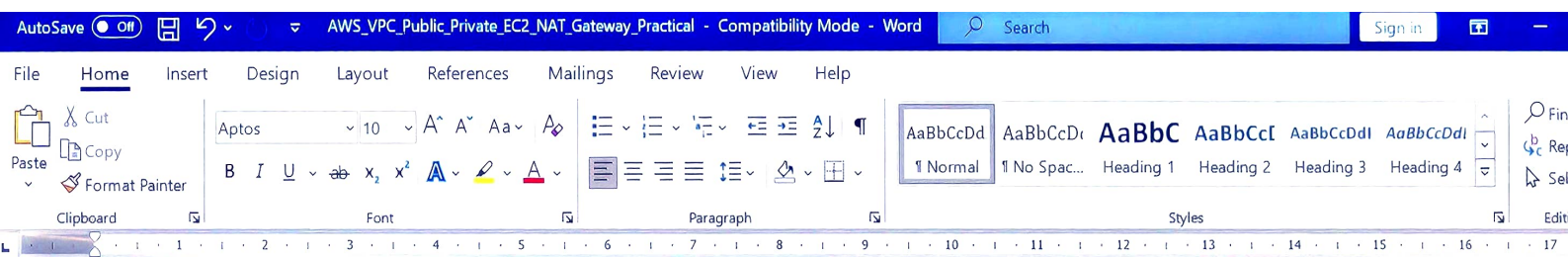
For Ubuntu, run:

```
sudo apt update
```

```
sudo apt upgrade -y
```

The exact package manager depends on the AMI you selected.





Step 19 — Install MySQL Server on Private EC2

For Amazon Linux 2023, MySQL package availability can vary by repository. First search available packages:

```
sudo dnf search mysql-server
```

If the MySQL server package is available, install it:

```
sudo dnf install -y mysql-server
```

Start and enable the service:

```
sudo systemctl enable --now mysqld
```

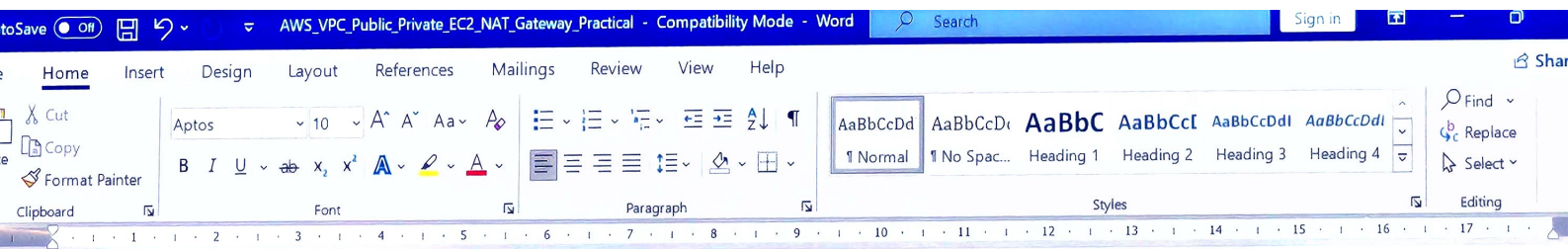
Verify:

```
sudo systemctl status mysqld  
mysql --version
```

If your chosen AMI uses MariaDB instead, the package/service name may differ. For a lab where MySQL specifically required, verify the package provided by the selected AMI repository before proceeding.

7. Optional — Verify MySQL Is Running





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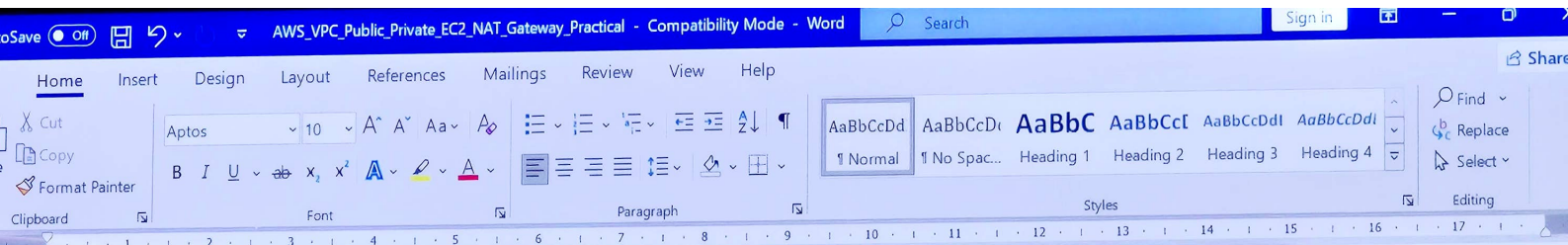
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```
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mysql --version
```

If your chosen AMI uses MariaDB instead, the package/service name may differ. For a lab where MySQL specifically is required, verify the package provided by the selected AMI repository before proceeding.

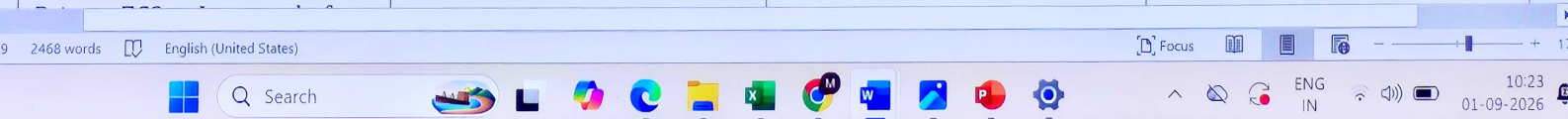
7. Optional — Verify MySQL Is Running

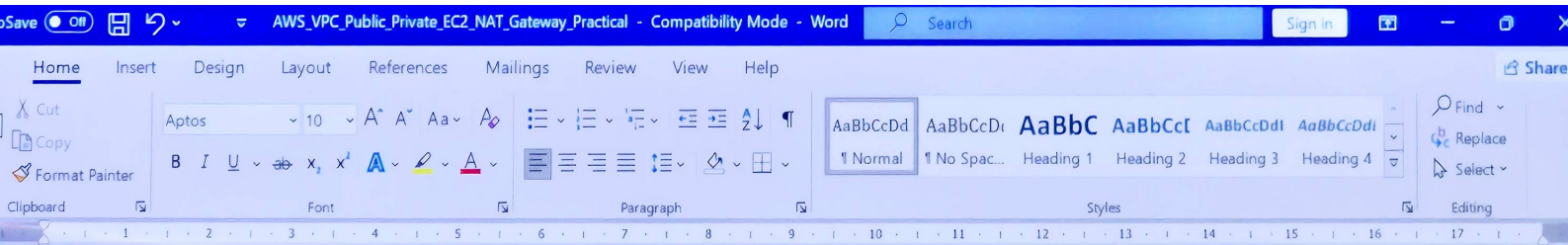
```
sudo ss -lntcp | grep 3306
```

This confirms whether a database process is listening on TCP port 3306. Do not expose port 3306 to the public Internet.

8. End-to-End Connectivity Test

Test	Expected Result	Why	Status
Laptop → Public EC2 :22	Success	Public subnet + public IP + IGW + SG rule	☐
Laptop → Private EC2 directly	Fail	No public IP / no direct inbound Internet path	☐
Public EC2 → Private EC2 :22	Success	Private SG allows SSH from Public-EC2-SG	☐

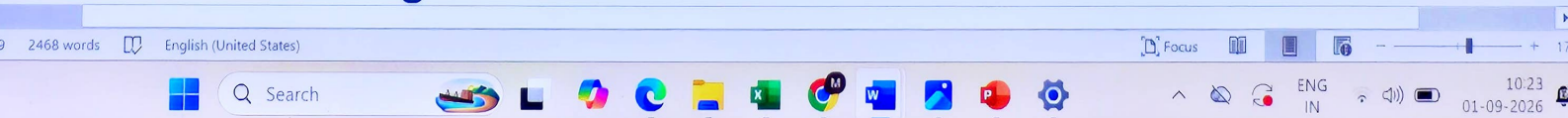


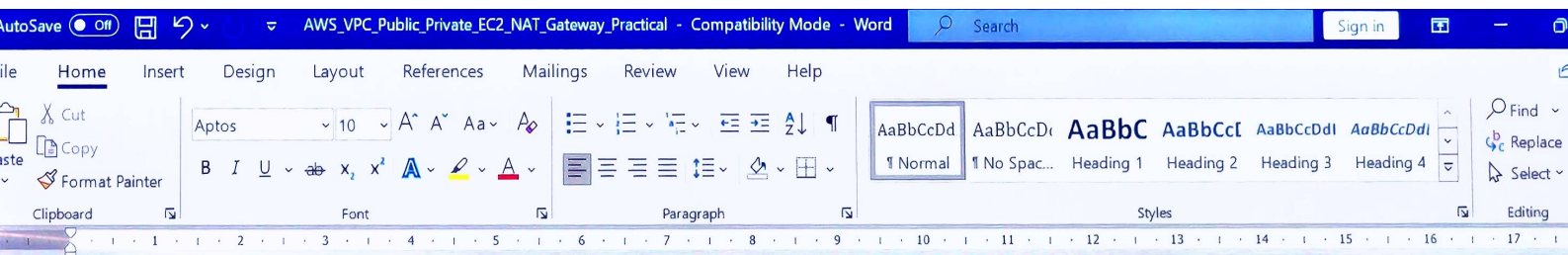


8. End-to-End Connectivity Test

Test	Expected Result	Why	Status
Laptop → Public EC2 :22	Success	Public subnet + public IP + IGW + SG rule	☐
Laptop → Private EC2 directly	Fail	No public IP / no direct inbound Internet path	☐
Public EC2 → Private EC2 :22	Success	Private SG allows SSH from Public-EC2-SG	☐
Private EC2 → Internet before NAT	Fail	No default Internet route	☐
Private EC2 → Internet after NAT	Success	Private default route points to NAT	☐
Private EC2 OS update	Success	NAT provides outbound Internet	☐
Install MySQL on Private EC2	Success	Package repository reachable through NAT	☐

9. Troubleshooting Guide





- Delete unused NAT Gateways and release unused Elastic IPs to avoid unnecessary charges.

11. Cleanup — Avoid AWS Charges

88. Terminate both EC2 instances.
89. Delete the NAT Gateway.
90. Release the Elastic IP associated with the NAT Gateway after the NAT Gateway is deleted.
91. Delete custom route tables after removing subnet associations.
92. Delete the custom NACL after removing subnet associations.
93. Detach and delete the Internet Gateway.
94. Delete both subnets.
95. Delete the VPC.
96. Delete the unused key pair from EC2 if it was created only for this lab.

Important: NAT Gateway is a paid AWS resource. Do not leave it running after the practical unless you intentionally need it.

